

```
mysql> INSERT INTO bank_transactions VALUES
-> (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05',NULL),
-> (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06',NULL),
-> (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08',NULL),
-> (104,'Anil','Vizag','Deposit',8000,'2024-01-10',NULL),
-> (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11',NULL),
-> (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12',NULL),
-> (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13',NULL),
-> (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14',NULL),
-> (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15',NULL),
-> (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16',NULL);
Query OK, 10 rows affected (0.02 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO bank_transactions
-> VALUES
-> (111,'Venu','Vizag','Deposit',7000,'2024-01-18',NULL);
Query OK, 1 row affected (0.02 sec)
```

```
mysql> UPDATE bank_transactions
-> SET amount = 5000
-> WHERE txn_id = 105;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> DELETE FROM bank_transactions
-> WHERE txn_id = 111;
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT * FROM bank_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

10 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM bank_transactions
-> WHERE transaction_type='Deposit';
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

6 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM bank_transactions
-> ORDER BY amount DESC;

+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | NULL |
+-----+-----+-----+-----+-----+-----+-----+

10 rows in set (0.00 sec)

mysql> CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'audit123';
ERROR 1396 (HY000): Operation CREATE USER failed for 'Auditor1'@'localhost'
mysql> CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'manager123';
ERROR 1396 (HY000): Operation CREATE USER failed for 'BranchManager'@'localhost'
mysql> GRANT SELECT
-> ON BankManagementDB.bank_transactions
-> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT ALL PRIVILEGES
-> ON BankManagementDB.bank_transactions
-> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> GRANT ALL PRIVILEGES
-> ON BankManagementDB.bank_transactions
-> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.03 sec)

mysql> REVOKE SELECT
-> ON BankManagementDB.bank_transactions
-> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql> REVOKE ALL PRIVILEGES
-> ON BankManagementDB.bank_transactions
-> FROM 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
-> SET amount = 6000
-> WHERE txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE bank_transactions
    -> SET amount = 99999
    -> WHERE txn_id = 102;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
    -> SET amount = 99999
    -> WHERE txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
    -> SET amount = 7000
    -> WHERE txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
    -> SET amount = 9000
    -> WHERE txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO SP1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

```

```

mysql> SELECT txn_id, customer_name, amount
    -> FROM bank_transactions
    -> WHERE txn_id IN (101,102);

```

txn_id	customer_name	amount
101	Ravi	7000.00
102	Sita	2000.00

2 rows in set (0.00 sec)

```

mysql> SELECT * FROM bank_transactions;

```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	7000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

10 rows in set (0.00 sec)

2520080012 dbms practical

```
mysql> CREATE DATABASE BankManagementDB;
Query OK, 1 row affected (0.03 sec)

mysql> USE BankManagementDB;
Database changed
mysql> CREATE TABLE bank_transactions (
  ->     txn_id INT PRIMARY KEY,
  ->     customer_name VARCHAR(50),
  ->     branch_name VARCHAR(50),
  ->     transaction_type VARCHAR(20),
  ->     amount DECIMAL(10,2),
  ->     transaction_date DATE
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> ALTER TABLE bank_transactions
  -> ADD account_no VARCHAR(20);
Query OK, 0 rows affected (0.11 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE bank_transactions
  -> MODIFY customer_name VARCHAR(100);
Query OK, 0 rows affected (0.12 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> RENAME TABLE bank_transactions
  -> TO customer_transactions;
Query OK, 0 rows affected (0.04 sec)

mysql> RENAME TABLE customer_transactions
  -> TO bank_transactions;
Query OK, 0 rows affected (0.05 sec)
```